📘 Phase 4 – Database Design

Chapter 1 – Database Overview

1.1 Purpose

The purpose of this document is to define the complete database structure of the **Gold Trading Automation Platform (GTAP)**. It describes all collections, fields, relationships, validation rules, and indexing strategy used by the system.

1.2 Database Technology

Property	Value
Database	MongoDB
ODM	Mongoose
Database Type	NoSQL Document Database
Data Format	BSON
Backend	Node.js + Express.js

1.3 Why MongoDB?

MongoDB has been selected because it offers:

- Flexible schema
- High performance
- Easy scalability
- Fast CRUD operations
- Native support with Node.js (Mongoose)
- Excellent handling of JSON-like documents

Chapter 2 – Database Collections

हमारे सिस्टम में Version 1.0 के लिए निम्न Collections होंगी:

Collection	Purpose
users	User authentication and profile
trades	All executed trades
bot_settings	Trading configuration
trade_logs	Bot activity logs
reports	Daily, weekly and monthly reports
sessions	Active login sessions

📘 Chapter 3 – Collection Relationships

```
User
|
├──────────────┐
|         |
▼         ▼
Trades    Bot Settings
|
▼
Trade Logs
|
▼
Reports
```

Relationship Summary

- One User → Many Trades
- One User → One Bot Setting
- One Trade → Many Logs
- Reports generated from Trades

📘 Chapter 4 – Users Collection

Collection Name:

users

Fields

Field	Type	Required	Description
_id	ObjectId	Yes	MongoDB ID
fullName	String	Yes	User Name
email	String	Yes	Login Email
password	String	Yes	Hashed Password
role	String	Yes	Trader/Admin
isActive	Boolean	Yes	Account Status
createdAt	Date	Yes	Creation Date
updatedAt	Date	Yes	Last Update

Validation Rules

- Email must be unique
- Password must be hashed
- Role default = Trader
- Active default = true

Indexes

- email (Unique)
- role

Sample Document

```
{
 "_id": "ObjectId",
 "fullName": "Vinit Kushwaha",
 "email": "vinit@example.com",
 "password": "hashed_password",
 "role": "Trader",
 "isActive": true,
 "createdAt": "2026-07-05T10:00:00Z",
 "updatedAt": "2026-07-05T10:00:00Z"
}
```

🎯 Phase 4 Progress

आज हमने Database Design की Foundation रख दी।

✅ Chapter 1 – Database Overview

✅ Chapter 2 – Collections Overview

✅ Chapter 3 – Collection Relationships

✅ Chapter 4 – Users Collection

- ✅ Chapter 6 – Bot Settings Collection
- ✅ Chapter 7 – Trade Logs Collection
- ✅ Chapter 8 – Reports Collection

📘 Chapter 5 – Trades Collection

Collection Name:

trades

Purpose

यह Collection सिस्टम द्वारा Execute किए गए प्रत्येक Trade को Store करेगी।

Fields

Field	Type	Required	Description
_id	ObjectId	Yes	MongoDB ID
userId	ObjectId	Yes	Reference to User
mt5Ticket	Number	Yes	MT5 Ticket Number
symbol	String	Yes	XAUUSD
orderType	String	Yes	BUY / SELL
lotSize	Number	Yes	Trade Lot Size
entryPrice	Number	Yes	Entry Price

exitPrice	Number	No	Exit Price
stopLoss	Number	Yes	Stop Loss
takeProfit	Number	Yes	Take Profit
riskReward	String	Yes	1:2
strategy	String	Yes	EMA11-RSI14-Fractal
status	String	Yes	OPEN / CLOSED
profitLoss	Number	No	Profit or Loss
openTime	Date	Yes	Trade Open Time
closeTime	Date	No	Trade Close Time
duration	Number	No	Trade Duration (seconds)
closeReason	String	No	TP / SL / Manual
createdAt	Date	Yes	Record Created
updatedAt	Date	Yes	Record Updated

Indexes

- userId
- mt5Ticket (Unique)
- status
- openTime

Sample Document

{
"userId": "ObjectId",
"mt5Ticket": 456789123,
"symbol": "XAUUSD",
"orderType": "BUY",
"lotSize": 0.2,
"entryPrice": 3352.40,
"exitPrice": 3358.60,
"stopLoss": 3349.90,
"takeProfit": 3357.40,

"riskReward": "1:2",
"strategy": "EMA11-RSI14-Fractal",
"status": "CLOSED",
"profitLoss": 124.80,
"openTime": "2026-07-06T10:15:00Z",
"closeTime": "2026-07-06T10:21:00Z",
"duration": 360,
"closeReason": "TP"
}

📘 Chapter 6 – Bot Settings Collection

Collection Name:

bot_settings

Purpose

यह Collection Bot की सभी Configurations Store करेगी।

Fields

Field	Type
userId	ObjectId
broker	String
accountNumber	String
server	String
symbol	String
timeframe	String
emaPeriod	Number
rsiPeriod	Number
fractalPeriod	Number
lotSize	Number
maxTrades	Number

stopLossBuffer	Number
riskReward	String
trailingSL	Boolean
telegramEnabled	Boolean
botStatus	Boolean
createdAt	Date
updatedAt	Date

📘 Chapter 7 – Trade Logs Collection

Collection Name:

trade_logs

Purpose

Bot द्वारा होने वाली सभी Activities Log होंगी।

Fields

Field	Type
userId	ObjectId
tradeId	ObjectId
logType	String
message	String
level	String
createdAt	Date

Example Logs

- Bot Started

- MT5 Connected
- Buy Executed
- Sell Executed
- SL Hit
- TP Hit
- Connection Lost
- Connection Restored

📘 Chapter 8 – Reports Collection

Collection Name:

reports

Purpose

Dashboard Reports के लिए Summary Data Store होगी।

Fields

Field	Type
userId	ObjectId
reportDate	Date
totalTrades	Number
winningTrades	Number
losingTrades	Number
winRate	Number
totalProfit	Number
totalLoss	Number
netProfit	Number
drawdown	Number
averageRR	Number

createdAt	Date

Report Types

- Daily
- Weekly
- Monthly

📊 Database Relationship (Updated)

```
User
|
├────────────────┐
|         |
▼         ▼
Trades    Bot Settings
|
▼
Trade Logs
|
▼
Reports
```

📘 Chapter 9 – Database Indexing Strategy

9.1 Purpose

Indexes improve database performance by reducing the time required to search, sort, and retrieve records.

9.2 Users Collection

Field	Index Type	Reason
email	Unique	Fast Login & Prevent Duplicate Users

role	Normal	Future Role Filtering

9.3 Trades Collection

Field	Index Type	Reason
mt5Ticket	Unique	One Trade = One MT5 Ticket
userId	Normal	User-wise Trade Search
status	Normal	Open/Closed Trade Filter
openTime	Descending	Latest Trades First
symbol	Normal	Filter by Trading Pair

9.4 Reports Collection

Field	Index Type	Reason
userId	Normal	User Reports
reportDate	Descending	Latest Reports
reportType	Normal	Daily/Weekly/Monthly

9.5 Trade Logs Collection

Field	Index Type	Reason
tradeId	Normal	Logs by Trade
createdAt	Descending	Latest Logs

📘 Chapter 10 – Database Validation Rules

Users Collection

- Email must be unique.
- Email must be a valid email format.
- Password must be hashed using bcrypt.
- Role can only be:
  - Trader
  - Admin (Future)
- Full Name cannot be empty.

Trades Collection

- Symbol = XAUUSD
- Order Type = BUY or SELL
- Lot Size > 0
- Stop Loss > 0
- Take Profit > 0
- Risk Reward = 1:2
- Status = OPEN or CLOSED
- MT5 Ticket must be unique.

Bot Settings

- EMA = 11
- RSI = 14
- Max Trades = 5
- Timeframe = M1
- Trailing Stop = Boolean

Reports

- Win Rate = 0–100%
- Drawdown ≥ 0
- Total Trades ≥ 0

📘 Chapter 11 – Backup & Recovery Strategy

Backup Policy

The system shall create:

- Daily Database Backup
- Weekly Full Backup
- Monthly Archive Backup

Backup Storage

- VPS Local Storage
- Cloud Backup (Future)

Recovery Plan

If the VPS crashes:

1. Restore MongoDB Backup.
2. Restore Environment Variables.
3. Restart Node.js using PM2.
4. Restart Python Trading Engine.
5. Verify MT5 Connection.
6. Resume Trading.

Data Protection

- Automated Backup
- Backup Verification
- Recovery Testing

📘 Chapter 12 – Database Best Practices

Naming Convention

Collections:

users
trades
reports
bot_settings
trade_logs
sessions

Fields:

camelCase

Examples:

entryPrice

takeProfit

createdAt

updatedAt

Security

- Never store plain text passwords.
- Use Environment Variables for database credentials.
- Restrict database access.
- Enable authentication in MongoDB.

Performance

- Use indexes only where needed.
- Avoid duplicate data.
- Store reports separately for faster dashboard loading.

Scalability

The database should support:

- Multiple Users
- Multiple Trading Accounts
- Multiple Strategies
- Additional Brokers

without major redesign.